

EDUCATION

PhD. Student : University of Nebraska-Lincoln

Aug 2023 - Present

Subject : Physics | Advisor : Dr. Ken Bloom | Chancellor's Fellowship Awardee

Bachelor of Science : Washington College

Jun 2020 - May 2023

Majors : Physics and Mathematics | Phi Beta Kappa, John S. Toll Physics Award

RESEARCH EXPERIENCE

Effective Field Theory Analysis of Top Processes

Jan 2025 - Present

University of Nebraska-Lincoln

URA Visiting Scholars Award Fall 2026 Recipient

- Implementing ML-based event categorization to improve bounds on quantities of interest.
- Wrote TopEFT processor for migration of analysis to Coffea 2024 hosted at the UNL Tier-3 computing center.

Dask Client for Analysis Grand Challenge (AGC)

Jan 2024 - Aug 2024

University of Nebraska-Lincoln

- Tested and benchmarked Dask client performance in 2024 AGC pipeline as part of 200 gbps challenge.
- Created sample notebook for 2024 AGC pipeline with Dask and Coffea 2024 for ATLAS use.
- Links to : [Presentation Slides](#)

Trigger Studies for di-Higgs processes

Oct 2022 - Jul 2023

Washington College | Advisor : Dr. Suyog Shrestha

US ATLAS SUPER 2023 grant recipient

- Studied trigger efficiency in search for 0-lepton di-Higgs processes. Links to : [Presentation Slides](#)
- Set up new particle physics lab suited for computer-based physics analysis @ Washington College. Configured, benchmarked, and tested software and hardware for analysis.

Exploring WIMP Dark Matter

Mar 2022 - May 2023

Washington College | Advisor : Dr. Digesh Raut

Cater Society of Junior Fellows, J.S. Toll Science Fellows grant recipient

- Calculated DM relic abundance with a freeze-out model of WIMP dark matter.
- Talk at AstroPhilly 2022 Symposium. Links to : [Presentation Slides](#), [Poster](#)

A New NanoAOD Era for CMS OpenData

May 2022 - Aug 2022

Fermilab LPC, University of Nebraska-Lincoln | Advisors : Dr. Nick Smith, Dr. Oksana Shadura

USCMS PURSUE Internship

- Created 2015 NanoAOD era for CMS OpenData and converted pre-2016 CMS OpenData to NanoAOD.
- Links to : [GitHub Repository](#), [Presentation Slides](#)

Background Modeling for ATLAS Di-Higgs Search

May 2021 - Jul 2021

Washington College | Advisor : Dr. Suyog Shrestha

J.S. Toll Science Fellows grant recipient

- Simulated and validated top-quark background signals for di-Higgs processes. Links to : [Poster](#)

TEACHING EXPERIENCE

Facilitator for USCMS PURSUE Internship 2026 @ Fermilab LPC 26 May - 5 June 2026

- Facilitated 2-week software training sessions for 2026 USCMS PURSUE Interns.
- Organized material for and conducted Git training sessions to equip students with skills to use Git tools from the command line. Links to : [Git Novice Fork](#), [Indico Page](#)

Graduate Teaching Assistant | University of Nebraska-Lincoln Aug 2023 - May 2025

- Led recitation and physics help sessions for undergraduate physics while taking graduate classes.

Particle Physics Outreach in Bengaluru, India Jan 2023

Washington College | Cater Society of Junior Fellows grant recipient

- Extended PURSUE 2022 research to create particle physics outreach program for high school students in Bengaluru, India. Used python instead of ROOT and Linux for accessibility.
- Students analyzed Z boson decay to muon-antimuon pairs and plotted resonance using CMS OpenData and python.
- Links to : [Lecture Slides \(1\)](#), [Lecture Slides \(2\)](#), [Project code+plots generated by students](#)

Physics and Mathematics Tutor | Washington College Oct 2021 - May 2023

- Bridge course design : Designed a math bridge course to help students with weak math backgrounds adjust to intro-level physics courses.
- Tutoring : Tutored physics and mathematics at the Washington College Quantitative Skills Center.

SKILLS

Programming : Python, ROOT, C++, Wolfram Mathematica, Unix+Bash/Shell, Virtual Machines with Docker, Git tools, Coffea AF , Android Studio.

Scripting/Typesetting : HTML/CSS, Latex and Beamer, Markdown

Research : Scientific modeling, data collection + visualization, statistical analysis.

Scientific Writing : Grant and Report writing, Literature review.

SERVICE

Treasurer, Physics and Astronomy Graduate Students' Association (PAGSA) April 2026 - Present

- Working to organize events for Physics and Astronomy graduate student engagement.

STEM Sisters @ Washington College Feb 2022 - May 2023

- Founded interest group for women+gender minorities in STEM at Washington College.
- Conducted weekly python, HTML, and tech literacy classes for women+gender minority students.

Improving Science Education @ Public Schools in Bengaluru, India Jul 2022 - Oct 2022

- Worked with non-profit Curiouscity to improve science education in public schools in Bengaluru, India.
- Designed cost-effective experiments that can be repeated by students at home to explore scientific concepts.
- Made accessible and lucid educational posters for scientific literacy at school.